Tier 1 Lab: Hooke's Law and Harmonic Oscillation

Exploring how well springs follow Hooke's Law; Exploring different regimes of harmonic oscillation

Objectives

- Introduce students to the iOLab device
- Introduce students to recording an experiment in a lab notebook
- Give students practice using varied software to interpolate, plot, and analyze data
- Give students practice applying linear and nonlinear fits to data

Overview

This lab is divided into several sections. You will first gain familiarity with the iOLab by measuring the weight set mass. The rest of the lab explores various aspects of Hooke's Law. You will measure the relationship between force and extension for springs on the tabletop and see how well they can be described by Hooke's Law ($F = -kx$) by measuring force as a function of displacement for one spring. After confirming the Hookean behavior of springs, you will explore the dynamical behavior of the spring-mass system.

For this experiment, we will use a compact data acquisition system called iOLab, which enables you to measure force, position, velocity and acceleration in real time. After data collection, you will use software to sharpen your data and error analysis skills.

What to Turn In

The group as a whole will submit a single copy of a lab report pdf on Gradescope, with plots. A separate bCourses submission is required for every Python or Excel file used in the analysis**.** Lab report expectations and specifics on the sections will be provided separately**.**

Online notebooks will be used to record the setup with pictures, experimental methodology, and data with notations contemporaneously. Each group can choose an online lab notebook format, such as a Google doc. The notebook will be used as the source for lab report sections, so it should be complete and orderly.

iOLab data can be shared within the group using the online repository. Each data set should also be shared with the instructors' special class repository account. Indicate which data set is used in the lab report.

Equipment

Equipment setup pre labs are not always included in Tier 1 and 2 labs. Read through the lab manual and assemble the necessary equipment in advance, so you can perform the experiment during class time to attach the iOLab spring along a surface.

- **iOLab**
- **iOLab Software**
- **Dowel**
- **Protractor**
- **Mass Set**
- **Electronic Scale**
- **Long Spring**
- **Pressboard**
- **Lab Stand**

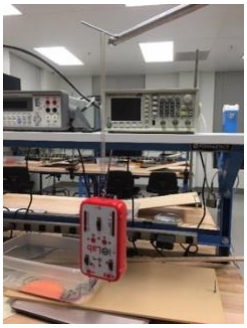
Experimental Setups

Setup along a surface (0°):

We will use the pressboard to create a firm, smooth surface and stable attachment. The iOLab spring fits snugly on the dowel but the spring should also be held securely in place on the dowel during data collection. The dowel itself should be held securely, as any movement shows up in the data.

The figures show the end-view (left) and side view (right) for the Hooke's law 0° case.

Setup for the iOLab 90° vertical harmonic oscillations:



The highest priority is a stable setup with sufficient vertical distance for the largest iOLab harmonic oscillation extent. The iOLab spring fits snugly on the dowel, which can be used as a support, and must be held securely in place on the dowel. The dowel should be held securely using a lab stand. In class, a lab stand should be used. Play with the setup to determine which dowel height works best for your setup. You should also consider having the spring sit closer to the lab stand end of the dowel to prevent additional motion or distortion due to flexing of the dowel itself.

The figure shows an example of the 90° vertical harmonic oscillator setup using a lab stand.

Setup for the iOLab 60° ramp oscillations (Figure 3):

For this section, the pressboard should be positioned over the lab stand rod.

The figure shows an example of the 60° ramp harmonic oscillator setup using a lab stand.

(Re-)Introduction to the iOLab

Starting with this experiment, we will more fully utilize the capabilities of the iOLab data acquisition system to take scientific measurement. Fig. 1 gives you an overview on the device and labels its various built-in sensors. It has one set of wheel sensor, accelerometer, gyroscope, force probe, barometer, magnetometer, microphone, thermometer, electrocardiogram, photodiode light sensor, voltmeter, voltage outputs and high-gain amplifier.

In this lab, we will use the iOLab's force and wheel sensors to first measure the weight set mass and then explore the spring's Hookean behavior, as well as to study harmonic oscillators. But before that, let's explore how to use the iOLab and its corresponding software to take measurement (also see "iOLab Help and Practice Recap" bCourses page).

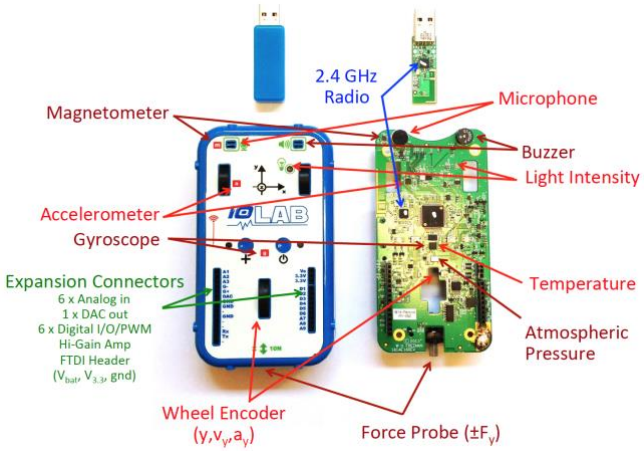


Figure 1. Design of the iOLab with all the important parts labelled. Copyright: www.iOLab.science

Before taking data

1. iOLabs connect to computers wirelessly. Plug the dongle (signal receiver) into the computer's USB port. Open the iOLab software and turn on your iOLab device. If nothing is paired, follow the instruction in the software to pair them.
2. ***Remove the clear tape from the back of the iOLab***. Do *not* place the dongle in the hole – the fit is too tight, and it can fall apart. Keep the dongle in the cardboard box.
3. Calibrate the sensors at the beginning. You might also need to do so when the batteries get changed.
4. In the software, select the sensors you need to use.
5. Re-zero the sensors before taking data. You'll do this when everything is ready: if the unit is flipped end-for-end vertically, so the force sensor orientation changes from the bottom to the top, for example, you'd need to re-zero.
6. When the experiment is ready to go, hit the "Record" button.

While taking data

1. While the data is being recorded, perform the experiment (initiate an oscillation or measure force-distance curve, etc).
2. If the values look odd, re-zero and recalibrate as needed.

After taking data

1. When the experiment is done, click the stop button.
2. If the data looks like what you want, click the folder button, find the last item and include a comment. Create a snapshot for the data with the same comment, which gets uploaded to the repository. This will allow you to more easily identify the data in the future.
3. Set the smoothing to 9 or an appropriate value before uploading (bottom left of plot).
4. For your good data, upload it to the online repository.
5. CSV files can be downloaded from the repository for analysis.

WARNING: Different iOLab sensors sample at different rates. The online repository aligns the time axes of different sensors automatically on the upload. Choose the highest Hz value for the upload.

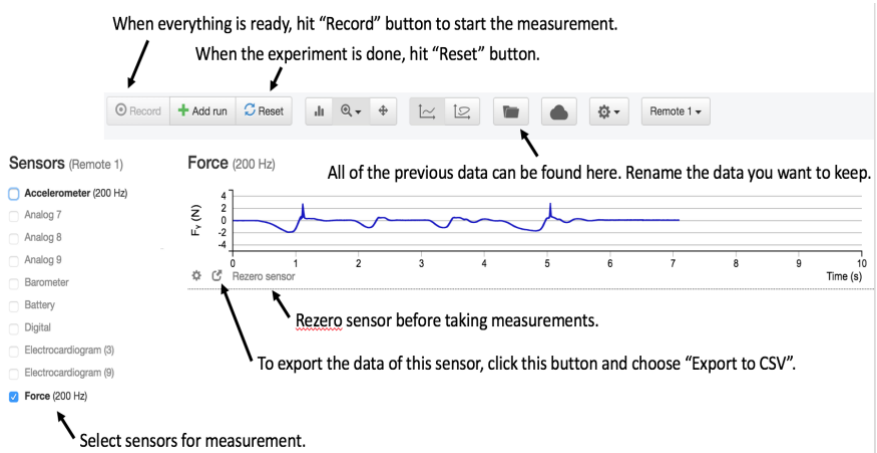


Figure 2. A brief introduction to the iOLab software operation. We will learn more of it as we go along.

Theory and background

Hooke's Law

When an object is stretched or compressed, it typically exerts a force opposing the change in length. For springs, this force is well-modeled by Hooke's Law, which states a spring exerts a force $F = -kx$ when stretched or compressed a distance x from its equilibrium length. The parameter k is a constant known as the spring constant. Because $F \propto x$, we say the force is linear in the extension of the spring.

It turns out that many real-world objects follow Hooke's Law when they are stretched or compressed by a small amount. Most objects, however, exhibit a nonlinear dependence on x when stretched or compressed by larger amounts. Rubber bands are an example of this: when you stretch a rubber band, there is a range of extensions for which $F \approx -kx$ for some constant value k. For very large or small extensions, however, the force deviates from this linear behavior.

Simple Harmonic Oscillation (SHO)

Let's consider a mass attached to a spring. From the equilibrium point, if the mass is displaced and released, it begins to oscillate in such a way that: $F = -ky(t) = ma_y(t) = my''(t)$. As a result, we get the differential equation

$$y''(t) + \omega_0^2 y(t) = 0,$$

where $\omega_0 = \sqrt{k/m}$ is the frequency of oscillation. This differential equation has the solution

$$y(t) = A\cos(\omega_0 t + \phi),$$

for some amplitude A and phase ϕ, and the motion is called simple harmonic oscillation. Notice the amplitude is constant for all times t. The velocity of the mass is

$$y'(t) = -A\omega_0 \sin(\omega_0 t + \phi),$$

while the acceleration is

$$a_y^{harmonic}(t) = y''(t) = -A\omega_0^2 \cos(\omega_0 t + \phi) \tag{1}$$

Damped Harmonic Oscillation with Velocity-Dependent Friction

Real spring-mass systems dissipate energy due to friction. Different kinds of friction have different dependence on velocity. A friction force proportional to velocity, $f_{viscous} = bv$, such as when a solid material moves inside fluid at a low speed is often encountered.

The differential equation from before becomes

$$y''(t) + 2\beta y'(t) + \omega_0^2 y(t) = 0,$$

where $\beta = b/2m$ is a constant. This differential equation from damped harmonic motion has the solution

$$y(t) = Ae^{-\beta t}\cos(\omega t + \phi),$$

where is new oscillation frequency is

$$\omega = \omega_0\sqrt{1 - \left(\frac{\beta}{\omega_0}\right)^2} = \sqrt{\omega_0^2 - \beta^2}.$$

The acceleration of the oscillating mass is

$$a_y^{damped}(t) = y''(t) = Ae^{-\beta t}[(\beta^2 - \omega^2)\cos(\omega t + \phi) + 2\beta\omega\sin(\omega t + \phi)] \quad (2)$$

The spring we will be using have $\beta/\omega_0 < 1$, so the motion is "under-damped". As the $e^{-\beta t}$ term shows, the amplitude of the mass oscillation decreases exponentially. Other values of β/ω_0 lead to qualitatively different behaviors, but those won't concern us for this lab.

Damped harmonic oscillation with constant friction

If a solid material is sliding on another solid material, the kinetic friction f is usually a constant and doesn't depend on the velocity. We get the differential equation

$$y''(t) + \omega_0^2 y(t) \pm \frac{f}{m} = 0,$$

where the plus sign is taken when the object moves in $+x$ direction, and the negative sign is taken when the object moves in $-x$ direction. Due to this $\pm$ sign, the solution of this equation is always a piecewise function. The easiest way to bypass this problem is to consider the "tuning points" during motion. Suppose the object starts statically at $y = -A_0$ and after half a cycle, stops momentarily at the first turning point $y = A_1$. From the work-energy principle, we know the reduction of elastic energy is due to the negative work of the friction, therefore,

$$\frac{1}{2}k(A_0^2 - A_1^2) = f(A_0 + A_1),$$

which leads to

$$A_0 - A_1 = 2f/k.$$

Similarly, for the next half cycle, we have

$$A_1 - A_2 = 2f/k$$

As you can see, the turning point positions A_0, A_1, A_2, A_3 ... form an arithmetic sequence. The turning point formula can be summarized as:

$$A_n - A_{n+1} = 2f/k, \quad (3)$$

where n indicates the nth peak and n+1 is the n+1th peak. This is the method we will use to identify damped harmonic oscillation with constant friction. [*For velocity-dependent friction, what kind of sequence will its turning points form? Notice the change of amplitude in y.*]

Part 0: Introduction to the iOLab: The Weight Set Mass

Overview of measurement:

The iOLab itself can be used as a mass scale. In this short section, finding the mass of the weight set will be used to gain familiarity with the iOLab force functionality. You should compare your iOLab measurements with the measurements given by the electronic scale.

Equipment

- **iOLab**
- **iOLab Software**
- **Python (or Excel)**
- **Mass Set**
- **Electronic Scale**
- **Online Lab Notebook**

Setup

Hold a calibrated iOLab steady with the force sensor pointing down, the eyehook screwed in, and no weight attached. Choose the force and accelerometer sensors. Zero the force sensor, then hit "Record" to start taking data. Attach the weight set, then hold the iOLab as steady as possible.

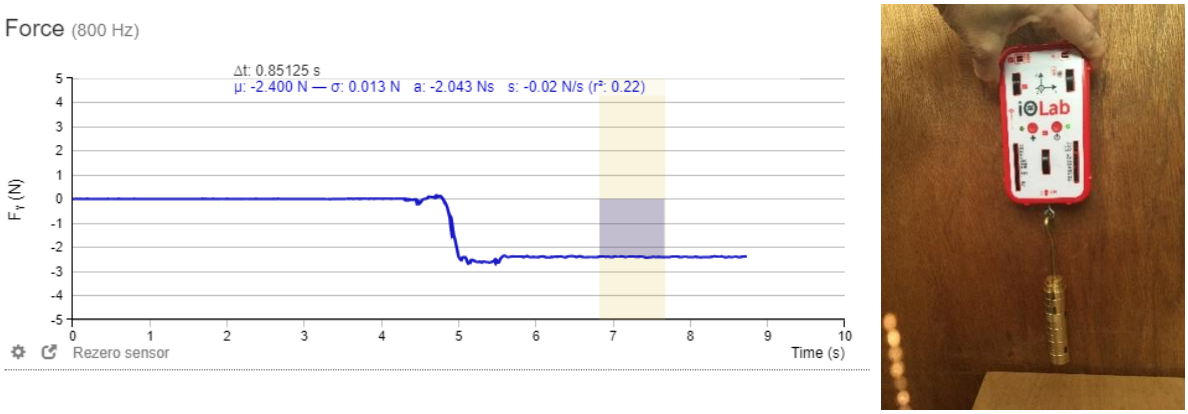


Figure 3. Force sensor data (left) and physical setup (right) for a 245.2 gm mass hanging from the IOLab

Remove the weight set while still recording, then rotate the iOLab once by 180 degrees, so the force is pointing upwards, then return to the original orientation and hold steady. Hit "Stop" to finish taking data.

Analysis:

Using the iOLab data analysis tools, evaluate the force in Newtons caused by adding the weight set. Data average along a quiet section for the best results.

Note: Tier 1 reports are not worksheets. You will need to decide where and how to address the following questions in your report.

- Do you obtain the correct weight result compared to the nominal value? Explain.
- Why is the accelerometer measurement not zero when the device is held steady? Explain.

Part 1: Procedure on Hooke's law (0° case)

Overview on Tier 1 data protocol

In this part, you will explore Hooke's law for a spring. Contrary to the previous labs, the procedure of Tier 1 experiments is more concise, so you need to fill in the gaps, and record the operations and data properly in the online lab notebook! Refer to the handout for the lab report sections to guide your lab notebook. Keep your lab notebook clean, but don't be afraid to take down random notes during experiments. Take pictures of your setup for the lab report and insert in the notebook. Writing down all the necessary information (including your thoughts) is good experimentalist practice and can help during data analysis!

Remember: Recording one data set is easy enough, but you need to convince yourself of your results before calling it good. This procedure is concise, but it is common to spend a few hours designing your experimental procedure, collecting good convincing data, and writing down the details in the lab notebook. Zero your force sensor every time before its use.

Equipment

- **iOLab**
- **Long Spring**
- **Mass Set**
- **Electronic Scale**
- **Pressboard**
- **Online Lab Notebook**

Tasks

Setup: You will be measuring force vs displacement for the silver spring on a horizontal surface (0°). Attach the iOLab to the spring under study. Place the iOLab on the pressboard then attach the spring's other end to the pressboard screw post (held securely).

Measure: Use the iOLab's force and wheel sensors to measure force vs displacement for the IOLab spring. Hints: a) re-zero the force sensor at the start, b) you may want to slowly roll the iOLab back and forth to gather more data in a single run, c) you may need to try multiple times (and develop some tricks) before you can get good-quality data.

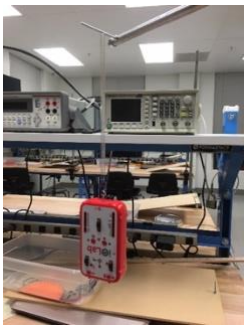
Export your data set: Set the data iOLab data smoothing to 9. Upload your best data set to the online repository. To analyze, download from the repository as a csv.

Data analysis:

- Select one spring extension (unloading or loading) from your data for further analysis.
- You will fit the force and distance data to Hooke's Law $F = -kx$ to derive the spring k constant, and the uncertainty in the spring constant α_k from the slope in simple linear least squares fit for the spring.
- Compare the fitted value to the accepted value (manufacturer value $= 12 \pm 10\%$) for the IOLab spring using an acceptance test.
- Graphs showing the raw data and fitted parameters, along with plots of resuduals, should be included in the relevant section of your report.

To fit the data, you need to export the data to a .csv file, import into Python (using the supplied interpolation and data fitting script) or Excel (using the supplied spreadsheet), select the region of interest, plot the data, and do the interpolation and Hooke's law model fitting.

Part 2: Procedure on Harmonic Oscillation (90º case)



Through the last part, we have established the Hookean behavior of springs. In this part, based on the Hookean behavior, you will explore harmonic oscillation of the spring-mass systems in different regimes.

Equipment

- **iOLab**
- **Long Spring**
- **Lab Stand**

Tasks

Setup: Hang the iOLab vertically from the dowel by the silver spring to make a harmonic oscillator. Try to develop tricks to make the oscillation as vertical as possible so the force is primarily in one axis and take several trial runs.

Measure:

- Use the iOLab's accelerometer and force sensors to measure 20 to 30 cycles of the harmonic oscillation of the iOLab-spring system. Try several times before identifying and exporting your best data set. Rezero the force sensor at the start or as necessary.
- In order to estimate the spring constant from this measurement, what other quantity do you need? Measure and record it. (Hint: you've already obtained it).

Export your data set: Set the iOLab data smoothing to 9. Upload your best data set to the online repository, then export the csv.

In-Class Analysis: Compare the accelerometer and force data that you have.

- Describe what they have in common and if they have any phase difference.
- For the quantity you measured in the second "Measure" bullet point, can you derive it without doing another measurement?

Data analysis: Your aim is to determine which regime of harmonic oscillation this experiment is in. Select a region of the acceleration data with many periods, then fit the following models to the same region and determine which provides the best fit.

Note: The spring constant depends only on the fit value of omega, which should be the same whether you use a simple harmonic fit or a damped harmonic fit. The point of this isn't just to calculate the spring constant but to determine which model fits the data best in each part. Assessing the accuracy of the models should primarily be done by looking at the plot of the residuals.

Fit 1: Simple harmonic oscillator (equation 1):

- Fit the acceleration data to the simple harmonic model with $a_y^{harmonic}(t)$ in the Theory section above to find the oscillation period using Python (or Excel) and derive the spring constant.
- Compare the spring constant (manufacturer value= 12 $\pm$ 10% N/m) with the value you found in 0º case, and comment on whether they are consistent or not.

Fit 2: Damped harmonic oscillation with velocity-dependent friction (equation 2):

- Fit the acceleration data to the damped harmonic oscillator with $a_y^{damped}(t)$ in the Theory using either Excel or Python. Find the oscillation period and the damping factor from this fit.

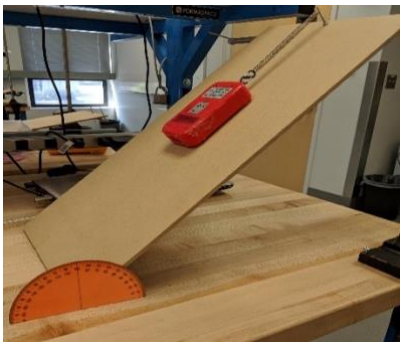
Additional analysis:

- Compare the oscillation frequencies from the simple and damped fits.
- Graphs of the acceleration vs. time with each fit should be included at the end of the report.

Part 3: Procedure on Harmonic Oscillation (60 ^{o} case)

hrough the last part, we have established the Hookean behavior of springs. In this part, based on the Hookean behavior, you will explore harmonic oscillation of the spring-mass systems in different regimes.

Equipment



- iOLab device
- Long spring
- Pressboard
- Protractor

Tasks

Setup: Put the iOLab-spring system (continue using the silver spring) on the adjustable-angle ramp. Set the ramp inclination angle to a 60° angle relative to horizontal. For this part, you'll be using the iOLab's wheel, accelerometer and force sensors to measure harmonic oscillation of the iOLab-spring system along the ramp.

Measure: You'll be measuring the harmonic oscillation of the iOLab-spring system along the ramp.

Don't overstretch the spring!

Before taking data, try the experiment several times with different oscillation amplitudes. You need sufficient periods of decreasing amplitude to ascertain which model applies. To get good enough data, obtain at least 10 oscillations. Equally importantly, aim for the final peak-to-peak amplitude to decrease to at least 1/7 of the original peak-to-peak amplitude. Be sure to examine the curves to ensure no artifacts appear in your data i.e. the curves should be smooth. One trick is to start the measurement before you pull the iOLab, in order to allow you zero the force probe and record the equilibrium position.

- Write in your notebook what the criteria for a good oscillation amplitude is and how many turns the oscillation usually lasts.
- Take data with a good oscillation amplitude, measuring force, wheel position, wheel velocity and wheel acceleration with the iOLab at the 60° inclination angle.

In-class Analysis: Pick a region of data in your best data set where the oscillations are large in each of force, position, velocity, and acceleration (you can use the wheel acceleration or the force data, whichever is quietest).

- Compare the periods to each other, e.g. by looking at the time difference between maxima.
- Are the relative phases consistent with what you'd expect from 1-diimensional kinematics and Newton's Laws?

Export Your Data Set: Set the iOLab data smoothing to 9. Upload your best data set to the online repository.

Data Analysis: Your aim is to determine which regime of harmonic oscillation this experiment is in. Selecting an adequate oscillation region for fitting is crucial; the initial starting peak should not be included but every oscillation thereafter should be, unless the amplitude is almost zero. The difference between Equations 2 and 3 fits becomes most obvious at the longest times. Do all fits on the same region.

Fit 1: Simple harmonic oscillator (equation 1):

- Fit the acceleration data to the simple harmonic model with $a_y^{harmonic}(t)$ in the Theory section above using either Python or an Excel workbook and find the oscillation period.

Fit 2: Damped harmonic oscillation with velocity-dependent friction (equation 2):

- Fit the acceleration data to the damped harmonic oscillator with $a_y^{damped}(t)$ in the Theory using either the Excel workbook or Python. Find the oscillation period and the damping factor from this fit.

Fit 3: Damped harmonic oscillation with constant friction (equation 3):

- Evaluate the peak amplitudes at the turning points for your data. Find the turning points directly from the wheel position or converting from the force data. Pick whichever sensor is the best and modify Equation 3 as needed.

Additional analysis:

- Which regime is the 60° harmonic oscillator in? Compare fits for the three regimes and state your interpretation of which model best describes your data and why.
- Graph of position, velocity, acceleration, and force data used for the "In-Class Analysis" should be included at the end of the report.
- Graphs of the acceleration vs. time with data and the fit for each regime should be included at the end of the report.

No $\tilde{\chi}^2$ analysis is required for this lab!

No error bars on plots of raw data.

Always include plots of residuals.

Capstone Project Ideas

In most of the following labs, you will find a section like this one. This section is usually about some additional interesting experiments which could become Capstone projects.

1. Continue to use the same setup in the last experiment, but double-side tape mass pieces to the iOLab to adjust the mass of it. Measure force, wheel position, wheel velocity and wheel acceleration with iOLab. Anticipate, measure and explain the changes in the data. How does the oscillation period and damping change? Does it make sense?

2. If you have time, you may find it interesting to explore **the behavior of a rubber band oscillator,** to see how it differs from simple harmonic motion.

Attach the rubber band to the iOLab and hang it from some stable point, like you did with the spring oscillator. Depending on the strength of your rubber band, may need to hang an additional mass from the iOLab to sufficiently stretch the rubber band so it can oscillate. If you don't know the mass of the object, you can use the iOLab as a scale by holding onto the iOLab and hanging the mass from the force probe.

Give the iOLab a tug as vertically as possible, and watch it oscillate. Record the acceleration data from the iOLab "accelerometer" and try fitting this data with a simple harmonic model like $a_y(t)$ in the Theory section above. Is the model a good match to the data?

Fit the simple harmonic model to the first few oscillations to get estimates for the amplitude, frequency, and decay constant. Use these to calculate the difference in kinetic energy after one period near the beginning of the oscillation (hint: at what times is all the energy kinetic?). Note this number.

Now use the iOLab to measure force vs displacement (via the wheel) for one quick cycle of the roughly the same amplitude. Estimate the area inside the hysteresis loop by looking at the parametric plot in the iOLab interface and convert this to an energy. Remember, this is the energy dissipated as heat in that one cycle.

How do your energy estimates compare? How should they compare?

Theory: Rubber bands and hysteresis

When you measure force vs displacement for a rubber band, you will notice a strange phenomenon: the rubber band exerts different amounts of force whether it's stretching away from equilibrium, or relaxing back towards equilibrium. Behavior like this is known as "hysteresis," and characteristically results in plots with loops like the one at right. For rubber bands, the asymmetric behavior is due to the polymers making up the rubber, which have interesting thermodynamic responses when stretched. (You'll learn about this in a Thermodynamics and Statistical Mechanics class later in the physics major.)

For the purposes of this lab, the important point is this: rubber bands heat up when they are stretched and cool down when they are relaxed. It therefore takes more energy to stretch the rubber bands because some of it gets dumped into heat. The total energy dissipated in one cycle (stretch and relax) is given by the area between the two curves. (Think about the work exerted in each direction.) Because of this behavior, a mass hanging from a rubber band also does not exhibit simple harmonic oscillation.

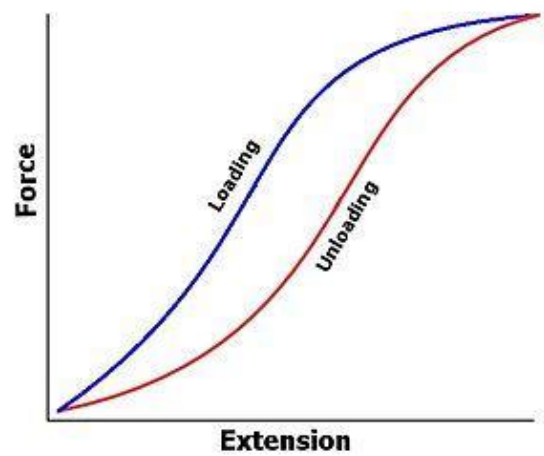


Figure: The force exerted by a rubber band is different whether it's loading (stretching away from equilibrium) or unloading (relaxing towards equilibrium).
Image credit: chubbyrevision.weebly.com

Procedure on Hooke's Law

You will use the iOLab's force and wheel sensors to **measure force vs displacement for two of your rubber bands**. *You should describe carefully the circumstances of each data set.*

- **Measure the curve several times for each one, recording the displacement amplitude and whether you are doing it quickly vs. slowly.** Record and print out your best data set (right-click on the plot in iOLab software and choose "Print"). Hint: It may be necessary to record multiple sets of data before you can make a conclusion.
- Compare the results when you do it quickly versus slowly for each rubber band, and when the amplitude is large or small. Did you notice any difference? Summarize your findings

Data analysis requirement: for **one "loading" curve** (e.g., one pull of the iOLab to stretch the rubber band), identify the region of extension that is approximately Hookean. Fit the data to Hooke's Law to derive the effective spring constant and use a statistical test to determine whether Hooke's Law is an appropriate model for this region of the data. Repeat this process for **one "unloading" curve of the same displacement**. Is the Hookean region the same? (*Note: You should describe carefully the circumstances of each data set, since the force curve is dependent on several things. In particular, how quickly did you pull or relax the rubber band, and did you do it immediately after other stretches/relaxes or after the rubber band had been left to rest for some time?)*